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Timestep-Conditioned Adapter Scaling for Multi-view Diffusion Texture Generation: Control Allocation and Baked Appearance

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ABSTRACT

Geometry-conditioned multi-view diffusion generates images that are baked onto 3D meshes, yet better image scores need not yield better textured assets. We study Timestep-Conditioned Adapter Scaling (TCAS) on an existing GeoTex residual interface, distinguishing requested scales, native-cap-adjusted applied scales, and state-dependent corrections. On 300 held-out objects, historical C3 improves foreground PSNR over Fixed Low by 0.522 dB but also increases the applied budget; Fixed High and Generic Linear have better means on all four image endpoints. At matched per-group applied-scale sums, the specified Middle-high minus mean(Early-high, Late-high) contrast is −0.0339 dB (95% interval $[-0.1172, +0.0514]$), with no detected middle-window advantage. Native-UV baking on ten compatible GeoTex objects finds a small Late-high unseen-view LPIPS reduction (−0.00265) alongside substantial reference-appearance error; a separate twenty-object GSO protocol does not detect the same benefit. Native-interface tests do not establish a transferable middle-window gain. A separate matched-input comparison under a recorded non-default protocol finds that the new geometry checkpoint remains below full v29 in FG-PSNR and FG-LPIPS, but does not isolate schedule effects. These findings support task-specific evaluation of control allocation and textured assets, not a universally preferred schedule or faithful material reproduction.

1. Introduction

Multi-view diffusion is an intermediate step in many 3D texturing systems: a reference image and geometry guide the generation of RGB views, which are then projected or baked onto a mesh. The graphics target is the textured asset viewed from new cameras. A useful inference control must therefore improve surface appearance under this complete workflow, rather than only a selected score on the generated images.

Geometry strength is a readily accessible control, but its numerical setting can obscure what an experiment changes. A temporal profile compared with a weaker constant also changes total strength. Existing wrapper caps may clip a requested profile differently at different injection sites. Even after matching the applied scalar sums, the correction tensors depend on the evolving hidden state. Consequently, a better score for a scheduled run does not by itself identify a benefit of its temporal placement.

The downstream endpoint adds a second distinction. Projection, coverage, resampling and agreement between incoming views affect the appearance of a baked texture. A generated-view improvement can remain small relative to reference-appearance error, and a texture-variation statistic can increase because of artifacts. These implementation and endpoint distinctions matter when deciding whether to adopt a geometry-control setting in a multi-view texturing system.

We investigate these questions using Timestep-Conditioned Adapter Scaling (TCAS), a requested scale profile on an existing GeoTex-Adapter residual path. The recorded C3 profile requests $(1.25, 2.50, 1.25)$ over the early, middle and late sampling windows. Conditioning schedules already exist: ControlNet exposes layer weights and temporal gates, and Scheduled Style Injection (SSI) schedules both attention-based style mixing and geometric ControlNet conditioning [40, 41, 31]. Our

research question concerns what controlled comparisons can establish for multi-view texturing after native clipping and asset construction. C3 is evaluated as a fixed operating point, without another schedule search or a claim of a new general scheduling principle.

The study links three questions: does C3 remain advantageous against competitive fixed and linear controls; what does moving the same high-scale segment establish at matched applied budgets; and how do these differences appear in actual UV-textured assets and other native conditioning interfaces? The answers revise the original interpretation of the profile. C3 improves over Fixed Low on the held-out GeoTex cohort, while stronger controls have better image-endpoint means. The specified equal-budget primary does not support a middle-window advantage. Baking reveals a small, object-dependent local benefit for a different allocation, together with substantial appearance errors and a separate-protocol non-detection. A full-native comparator exposes a larger deployment-quality gap, and a separate input cohort suggests texture-complexity-related response heterogeneity. Different conditioning interfaces also fail to establish a stable middle-window gain.

The supported contributions are:

(1) **An implementation-aware characterization of geometry-control allocation.** Requested, capped applied and realized residual quantities are distinguished, and competitive controls change the interpretation of the C3 gain. Matched per-group scalar sums test temporal placement without treating them as matched residual doses. The non-detected primary and separate pairwise effects constrain the earlier middle-window recommendation.

(2) **A measured connection between view-level control and baked appearance.** Native-UV meshes, matched-resolution ground-truth self-bakes and eleven unseen cameras quantify the size of allocation changes relative to persistent asset-appearance error. Positive image/bake rank associations coexist with changing point-estimate winners; a small local LPIPS gain does not establish faithful material reproduction or a benefit across the two bake protocols.

(3) **Evidence delimiting transfer across conditioning interfaces.** Native MV-Adapter and retained MVDiffusion studies test related allocation questions within their actual operators and camera contracts. Their small or endpoint-dependent responses limit schedule transfer and show why common profile labels cannot be assumed to encode a common intervention.

For graphics researchers, these findings affect both comparison design and deployment decisions: a schedule should be assessed against strong settings at the implemented control site, and its practical value should be measured on the resulting textured mesh. The evidence supports these bounded decisions without requiring a universally winning profile.

2. Related Work

2.1. 3D Texture Generation and Multi-view Diffusion

Single-image 3D generation is often decomposed into geometry generation and texture generation. Single-image-to-3D, multi-view diffusion, and sparse-view reconstruction methods have improved structural stability and view synthesis [1, 2, 3, 4, 5, 6, 7, 8]. Texture synthesis remains a distinct challenge for asset realism and usability [15]. Many systems generate RGB views with 2D diffusion and project or bake them onto a surface [9, 10, 11, 12]. Recent work also explores exemplar-based texture synthesis and diffusion-based procedural materials [16, 17].

Paint3D, MVPaint, Hunyuan3D 2.0, and MVPainter use geometry-conditioned multi-view generation in 3D texturing pipelines [18, 19, 20, 13]. MV-Adapter, MVDiffusion, and other synchronized multi-view methods address consistency through different adapter and denoising designs [21, 22, 23, 24]. These systems establish the graphics context for this work. The present evidence does not compare final baked-texture quality across those systems.

2.2. Adapter-Augmented Diffusion Models and Geometric Control

ControlNet and T2I-Adapter provide learned spatial-conditioning branches or feature injections, whose contribution can be weighted by an inference-time condition-strength scale [25, 26]. They establish conditioning strength as a practical control, but their injected tensors and scaling sites do not share a numerical dose with GeoTex. The studied GeoTex-Adapter path already exposes an inference-time residual scale and evaluates fixed requests. TCAS reuses that control point as a three-stage request over sampling progress, while recording how wrapper-group caps change the applied values and how adapter activations determine the resulting corrections. It adds no learned branch or schedule selector. IP-Adapter introduces image prompts through decoupled attention [27]; LoRA and related methods use scaled low-rank updates [28, 29, 30].

2.3. Diffusion Guidance Scheduling and Timestep Control

Diffusion denoising is often discussed in terms of coarse-to-fine evolution [32, 33]. Classifier-free guidance combines conditional and unconditional score predictions [34]; limiting it to an interval studies quality and diversity [14]. Adaptive Guidance instead schedules conditional/unconditional evaluations for inference efficiency [35]. These score-space and computation schedules differ from the geometry-residual intervention tested here.

Earlier official ControlNet code already exposes layer-dependent control weights, and Diffusers v0.21.0 documents temporal conditioning start/end gates [40, 41]. SSI is the closest combined layer/time scheduling precedent. It varies StyleID's query-mixing parameter [39] across decoder layers and denoising time, tests schedule direction and shape, and also schedules a separate geometry-conditioned ControlNet branch [31]. Thus, layer/time scheduling and geometry-conditioning schedules are established ideas. TCAS acts on the residual correction produced by GeoTex-Adapter from normal, depth, and foreground-mask conditions, with group-specific wrapper caps. In this 3D texturing setting, we measure how that intervention relates to generated-view agreement, reference-color errors, and baked appearance. SSI evaluates image style/content trade-offs. Our controlled question connects an implemented

geometry-residual allocation to generated views and native-UV assets. The differences in task, operator and endpoints motivate this evaluation; they do not establish novelty of temporal or layer scheduling itself.

2.4. Multi-view Adapter Interfaces

MV-Adapter uses a condition guider for camera or geometry maps together with decoupled spatial, multi-view, and optional image cross-attention layers [21]. MVDiffusion uses correspondence-aware attention modules to connect pixels across views [22]. These interfaces differ from the GeoTex-Adapter residual branch studied here. We therefore report each auxiliary intervention within its own tested implementation and cohort, without pooling metrics or interpreting it as an architecture-matched replication of TCAS. The native MV-Adapter condition means are summarized in Table 6; its specified schedule primary and the external-input native-interface primary are reported with the original Fresh C timing primary in Table 7. The two separate bake primaries remain in Table 5; complete secondary families remain in Supplementary Sections S3–S5.

3. Control Formulation and Study Design

3.1. Task Definition and Base Pipeline

We study inference-time geometry strength in an existing MVPainter-style multi-view texturing framework [13]. Reference conditioning follows the base image path; normal, depth and mask conditions enter GeoTex-Adapter. TCAS sets the requested multiplier on the existing residual branch. The existing wrapper caps that value before multiplication and addition to frozen diffusion U-Net features (Figure 1). The base U-Net, VAE and adapter weights remain frozen. Fresh C evaluates six generated RGB views; separately identified bake cohorts assess actual UV-textured meshes from eleven unseen cameras. Baking is a downstream evaluation, not a module introduced by TCAS.

During multi-view diffusion, the 3D geometry is rendered into normal maps, depth maps, and foreground masks, denoted as:

$$\mathbf{G} = \{G_{\text{normal}}, G_{\text{depth}}, G_{\text{mask}}\}. \tag{1}$$

The geometric adapter maps these conditions to a residual correction term and injects it into the intermediate hidden states of the UNet. For inference step j with normalized execution progress p_j, the adapter injection can be written as:

$$\mathbf{h}'_{j,\ell} = \mathbf{h}_{j,\ell} + s_{\text{app}}(p_j, \ell)\, A^{\ell}_{\phi}(\mathbf{h}_{j,\ell}, \mathbf{G}), \tag{2}$$

Here ℓ identifies a wrapper within one of the three injection groups; caps and requested profiles are assigned by group. The *requested scale* $s_{\text{req}}(p_j, \ell)$ is the value assigned by the profile; the *applied scale* is the value used by the wrapper,

$$s_{\text{app}}(p_j, \ell) = \min(s_{\text{req}}(p_j, \ell), c_\ell), \tag{3}$$

when the native cap c_ℓ is active, and otherwise equals the request. The resulting correction is $\mathbf{r}_{j,\ell} = s_{\text{app}}(p_j, \ell) A^{\ell}_{\phi}(\mathbf{h}_{j,\ell}, \mathbf{G})$; its *post-scale residual magnitude* is $m_{j,\ell} = \|\mathbf{r}_{j,\ell}\|_2$. Thus neither a requested nor an applied scale label alone determines the correction magnitude.

In the evaluated up-only Fresh C configuration, the GeoTex wrappers on UNet up blocks `up_0`, `up_1` and `up_2` are assigned to the deep, middle and shallow cap groups, respectively. These names identify the implementation's grouped injection points; they do not assert universal semantic roles for denoising stages. The corresponding Fresh C caps are 3.0, 3.5, and 0.8. The shallow requests in all three C3 phases, 1.25/2.50/1.25, are applied as 0.80/0.80/0.80; deep and middle retain the requested profile. These caps belong to the existing wrapper and are not introduced by TCAS. Their values and group names do not establish causal semantic roles. GFL and GFH also clip at the shallow cap. These values describe the evaluated Fresh C wrapper; row-level cap semantics for the historical 24-object probe are not recoverable. TCAS scales a geometry residual added to feature activations; it does not reweight conditional and unconditional score predictions as classifier-free guidance does.

3.2. Image-Space Metrics and Texture-Variation Diagnostics

The endpoints describe different outcomes. *Image-space agreement* refers to target-view PSNR, LPIPS, and SSIM measurements on the named image regions; it is not a measure of final 3D material quality. *Reference-color agreement* refers to color differences such as CIEDE2000 computed against the corresponding reference. *Texture variation* refers to statistics such as RGB standard deviation, Laplacian variance, gradient magnitude, and high-frequency energy; these may rise with detail or artifacts and are not fidelity measures. *Subjective preference* refers only to the submitted forced-choice study. *Baked 3D appearance* refers to rendered mesh outputs after texture projection or baking, while *PBR/material fidelity* additionally concerns appearance under the material and illumination model. These endpoints are not interchangeable. FG-SSIM, Edge-SSIM, and Crop-SSIM quantify structural similarity on the indicated image regions; PSNR measures pixel error; LPIPS compares learned image features [36, 37].

Laplacian variance, RGB standard deviation, and gradient magnitude can describe local high-frequency or color variation. For example, a foreground Laplacian statistic may be written as

$$M_{\text{var}} = \text{Var}(\text{Lap}(I_{\text{out}} \odot G_{\text{mask}})), \tag{4}$$

where G_{mask} is the foreground mask. A larger value is not necessarily better: detail, noise, repeated patterns, or artifacts can all increase local variation. We therefore do not interpret these proxies as texture fidelity or use them alone to claim that a schedule better preserves reference appearance.

3.3. TCAS Control Semantics and Comparison Design

TCAS replaces a globally fixed requested adapter scale with a piecewise profile over normalized sampling progress. The early/middle/late partition is an operational description, not a universal claim about the function of every denoising stage. C3's low–high–low profile is the recorded development choice evaluated in this study; it is not presented as an independently derived optimum.

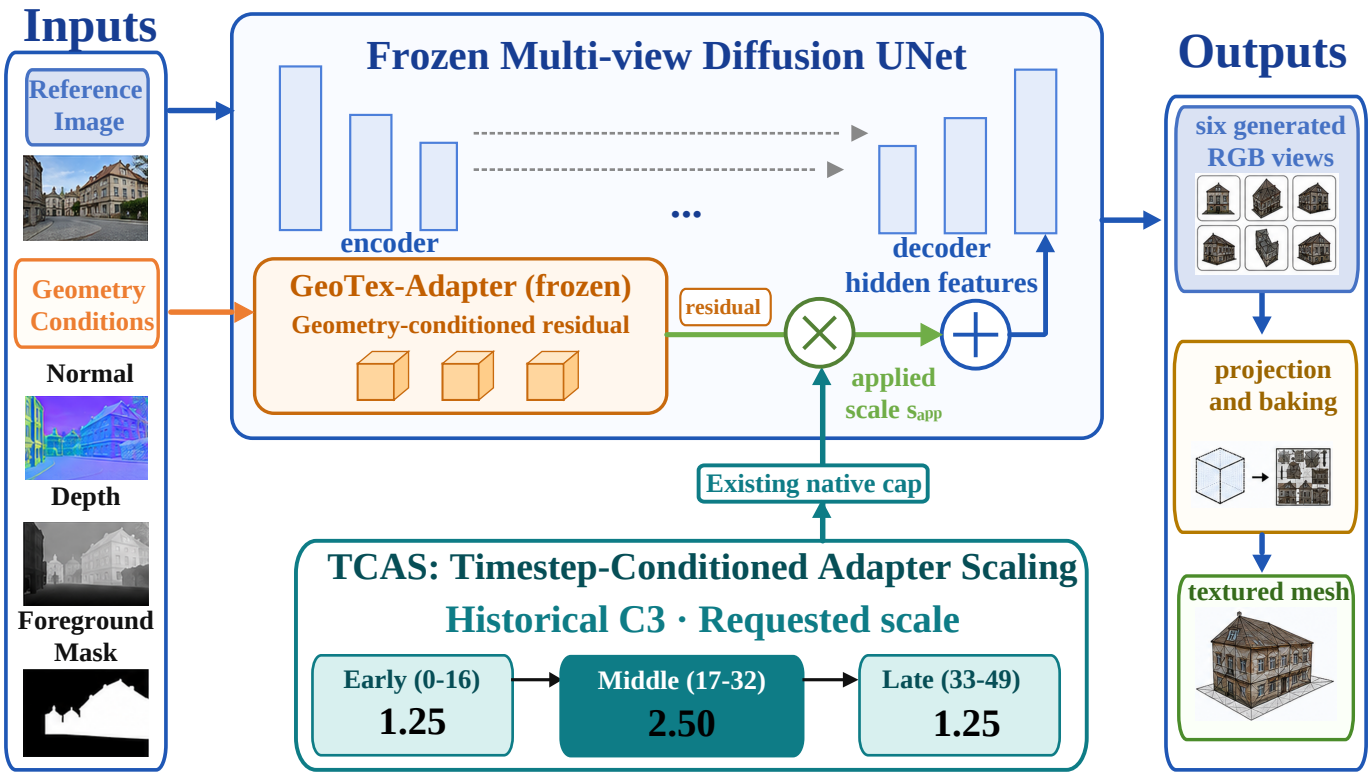


Fig. 1. Overview of the TCAS control interface in multi-view diffusion texturing. The frozen pretrained GeoTex-Adapter computes geometry-conditioned residual corrections from normal, depth and foreground-mask features, together with current hidden features. TCAS specifies the historical C3 requested scaling profile (1.25, 2.50, 1.25) over steps 0–16, 17–32 and 33–49. Existing native group-specific caps determine the applied scales before residual multiplication and addition to frozen U-Net features. In the evaluated Fresh C implementation, the deep, middle and shallow caps are 3.0, 3.5 and 0.8; shallow is clipped to 0.8 throughout. Six generated RGB views can subsequently be projected and baked onto a textured mesh. The image panels are illustrative and are not used as cohort evidence.

To avoid ambiguity caused by different raw timestep directions in different schedulers, $p_j = j/(T-1)$ denotes execution progress for inference step $j \in \{0, \dots, T-1\}$. Thus $p = 0$ is the first update from the initial noisy latent and $p = 1$ is the last denoising update before decoding. For $T = 50$, TCAS divides execution into steps 0–16, 17–32, and 33–49 (17/16/17 steps):

$$s_{\text{req}}(p) = \begin{cases} s_e, & 0 \le p < 1/3 \\ s_m, & 1/3 \le p < 2/3 \\ s_l, & 2/3 \le p \le 1 \end{cases}. \tag{5}$$

We call this variation across the 50 inference steps the *temporal allocation*. A *layer allocation* varies the requested scale across the three wrapper groups at a fixed step. For a schedule with applied values $s_{\text{app}}(j, \ell)$, its per-group applied-scale sum is $B_\ell = \sum_j s_{\text{app}}(j, \ell)$; the unweighted sum $B_{\text{nom}} = \sum_\ell B_\ell$ is a *nominal applied-scale budget*. It does not weight scales by adapter activations. We log post-scale correction summaries, including wrapper-level L_2 norm and mean absolute entry. The three wrapper groups operate on feature maps with different spatial dimensions, and the traces do not retain channel counts; raw L_2 values are therefore not used for cross-layer dose comparisons. The per-entry mean absolute correction reported in Supplementary Section S1 is a dimension-normalized descriptive summary, not an equal-dose intervention. Correction magnitudes also depend on the current hidden state and earlier denoising updates, so equal nominal budgets do not imply equal realized corrections.

The selected C3 configuration used in this paper is:

$$(s_e, s_m, s_l) = (1.25, 2.50, 1.25). \tag{6}$$

The low–high–low profile is a concrete inference-time intervention motivated by the submitted development analysis. Its stage interpretation is a task-specific hypothesis: the tested early, middle, and late allocations may produce different responses. Fresh C evaluates the recorded profile against matched controls but does not prove a universal function for any stage. TCAS adds no trained parameters and does not modify base-model or adapter weights; it replaces the requested scale during inference.

What C3 changes and what its comparisons identify. C3 requests (1.25, 2.50, 1.25) for each of the 50 steps across the three layer groups before native caps. In Fresh C, the applied deep and middle profiles retain those values; the shallow profile is clipped to 0.8 throughout. GFL requests 1.25 at every step and group, so its deep and middle values are 1.25 throughout and its shallow value is also clipped to 0.8. Their per-group applied-scale sums are therefore 82.5 versus 62.5 in deep and middle, and 40 for both in shallow. The C3–GFL contrast changes the middle-third deep/middle scale magnitude as well as its temporal allocation; it is not a timing-only comparison.

To identify a temporal-placement contrast, controls hold the per-layer applied values and nominal per-group sums fixed while moving the same values between time windows. A scale-strength contrast instead changes amplitude while preserving layer and temporal allocation. Matching requested values or applied-scale sums does not match the evolving residual corrections; raw L_2 values are also dimension-sensitive across wrapper groups and cannot establish cross-layer dose matching. The 300-object allocation comparison matches applied sums in each layer group and therefore estimates an allocation effect at a fixed scalar budget, not a residual-dose-independent mechanism. The historical-request sensitivity panel is reported separately.

Development origin and scope of C3. The submitted study records C3 = (1.25, 2.50, 1.25) as the selected profile from a 24-object development probe containing 17 uniform, layer-wise, and time-conditioned variants. The available historical records do not preserve the row-level outputs needed to reconstruct that ranking. We therefore treat C3 as the recorded operating point and make no optimality or uniqueness claim based on the original probe. Fresh C evaluates it without further schedule search.

3.4. Geometry Adapter and Inference Interface

The pretrained GeoTex-Adapter receives a five-channel input comprising a three-channel normal map, a one-channel depth map, and a one-channel foreground mask. Its multi-scale encoder produces residual features that are injected into the frozen UNet. TCAS acts on the requested multiplier along denoising progress; it does not retrain the adapter, the UNet, or the VAE and introduces no learned parameters.

Visual comparisons use recorded outputs with shared object framing and crops. They illustrate task behavior and appearance limitations; quantitative conclusions use the stated object-level metrics.

3.5. Study Cohorts, Sampling and Statistical Units

Table 1 summarizes the cohorts and protocols. The principal evaluation cohort, denoted Fresh C, contains 300 object identities disjoint from the historical schedule-selection/evaluation pool, the recorded 1,118-object training list and the earlier clean-v2 evaluation list. Its primary four-condition comparison uses one frozen retrained GeoTex checkpoint, shared inputs and paired object seeds. This study does not reproduce the historical adapter's reported +0.96 dB estimate; the original main checkpoint's exact identity remains unavailable. A separate historical-request retraining is reported as protocol sensitivity.

Within each comparison, object inputs, targets, initialization and endpoint code are shared. Fresh C uses seed 42 + `object_idx`, held constant across conditions for that object. Its capped residual groups have deep/middle/shallow limits of 3.0/3.5/0.8. MV-Adapter uses a different feature injection operator and numerical scale range; its native target cameras and actual default-path identity are checked separately. Scale labels are not common physical doses across interfaces.

Fresh C's controlled sampler differs from the complete base pipeline, which uses online image features, classifier-free guidance and 512-pixel views. Constant 1.0 under the GeoTex caps also differs from the unset-scale residual branch. We therefore name these controls by their executed configuration. The source records and sampling/default-path diagnostics are in Supplementary Sections S1 and S6.

FG-PSNR, LPIPS, SSIM and Edge-SSIM measure agreement on the named image regions. Baked evaluation renders actual exported meshes; unlit base color does not establish PBR or material fidelity. Fresh C image endpoints are evaluated once on each frozen 3×2 six-view composite. Native-interface and baked endpoints average their recorded per-view scores within object, with GSO sampling seeds also averaged within object. Paired intervals resample objects and are pointwise 95% intervals. Preserved primary, secondary and exploratory families use their stated Holm adjustments; the distinct families do not imply one overall experiment-wide error guarantee. Non-detection is not equivalence or non-inferiority. Complete comparison rules are supplied in Supplementary Sections S2–S5.

Reproducible analysis and presentation. Statistical analyses use NumPy/SciPy with the specified object resampling, seeds and original correction families; plots use Matplotlib. The accompanying scripts and source records document the calculations and inputs for numerical verification.

4. Experimental Results

4.1. Competitive Controls and the Historical C3 Profile

The internally specified Fresh C primary is C3 minus Fixed Low (GFL) FG-PSNR. The paired gain is +0.52 dB (95% interval [+0.40, +0.64]), with 206/300 objects favorable. All conditions use the same frozen retrained checkpoint and object inputs. This gain combines increased strength and temporal allocation: C3's deep/middle applied sums are 82.5 each, versus 62.5 for GFL, while both have shallow sum 40.

Competitive controls change the recommendation. Fixed High has better means than C3 on all four endpoints: C3 minus Fixed High is −0.748 dB FG-PSNR (pointwise interval [−0.885, −0.609]), with FG-SSIM difference −0.04553 ([−0.04937, −0.04167]). Generic Linear ramps deep/middle requests from 1.25 to 2.50 and shallow requests from 0.50 to 0.75 over the fifty steps. Its applied sums are (93.75,93.75,31.25), so it changes layer allocation and total budget as well as timing. It also has better means on all four; C3 minus Generic Linear is −0.593 dB ([−0.837, −0.351]). Its positive median of +0.074 dB and 155/300 favorable objects show why mean and object-level distributions must both be considered. Complete original paired endpoints and their comparison families are in Supplementary Section S2.

4.2. Temporal Allocation at Matched Applied-Scale Budgets

Early-, middle-, and late-high move the same sixteen high steps to updates 0–15, 17–32, and 34–49. Deep/middle low and high are 1.25 and 2.50; shallow stays at its 0.8 cap. Fixed Mean

Table 1. Studies and evidence scope. Cohorts may overlap; views and seeds remain nested within object.

Study / objects	Interface and sampling	Endpoint	Selection and scope
Fresh C / 300	Capped GeoTex residual; 256 px; 50 Euler steps; conditional-only	Six target-view FG metrics	UID-disjoint cohort; C3 primary; linear and matched-budget addenda
GeoTex bake / 10	Controlled generator; native UVs; six bake views	Eleven unseen unlit views	Compatibility-selected pilot; Late–Mean primary; two later controls exploratory
MV-Adapter / 50	Native condition-guider; 512 px; 50 steps; CFG 3	Six native-camera views	Input-selected; thirteen legal white targets retained
GSO / 100	Native MV-Adapter; 512 px; 50 steps; CFG 3; three seeds	All-six PSNR primary; non-reference LPIPS secondary	Source-ordered convenience cohort; external sensitivity
GSO bake / 20	Fixed GSO inputs; native UVs; three seeds	Eleven unseen unlit views	GT-only recalibration after initial outcomes; full cohort retained
Reference protocol / 50	New frozen GeoTex; native references; 512 px; CFG 2	Five nonreference views	Previously studied inputs; protocol/checkpoint sensitivity
Historical request / 276 slots	Retrained GeoTex; 1,039 training UIDs; one unknown replacement	Six-view composite; full-frame and FG	Exposed technical sensitivity; exact held-out UID coverage uncertified
MVDiffusion / 75	Partial correspondence-block scaling; depth unscaled	Recorded six-view endpoints	Retrospective third-interface panel; not exact GeoTex operator transfer

Table 2. Fresh C competitive-control means on the same 300 objects.

Control	Deep/middle sum	Shallow sum	FG-PSNR↑	FG-LPIPS↓	FG-SSIM↑	Edge-SSIM↑
Fixed Low	62.5	40	11.31	0.2051	0.3500	0.4728
Fixed High	125.0	40	12.58	0.1891	0.4201	0.4961
Fixed Mean	82.5	40	11.96	0.1978	0.3864	0.4843
Early-high	82.5	40	11.33	0.2045	0.3696	0.4765
C3 / Middle-high	82.5	40	11.83	0.1998	0.3746	0.4773
Late-high	82.5	40	12.39	0.1951	0.3788	0.4914
Generic Linear†	93.75	31.25	12.42	0.1862	0.4026	0.4880

Equal-budget rows (Fixed Mean, Early-high, C3/Middle-high and Late-high) share per-group sums of 82.5 for deep/middle and 40 for shallow. Generic Linear† is a same-cohort addendum with different sums (93.75, 31.25); it is not an equal-budget strategy.

uses 1.65 before the cap. Each allocation therefore has per-group sums $(82.5, 82.5, 40)$. All six conditions share the object inputs and seed, giving 1,800 paired object-condition rows.

The equal-budget addendum's internally specified primary question is Middle-high minus the objectwise average of Early-high and Late-high FG-PSNR. Its mean is −0.0339 dB (10,000-resample pointwise object-bootstrap interval $[-0.1172, +0.0514]$; two-sided 20,000-draw sign-flip $p = 0.4324$), with 119/300 objects positive. The median is −0.2296 dB. This result does not support a middle-window advantage and does not establish equivalence. It was specified before the addendum's new allocations, after earlier Fresh C outcomes were known; it is not an independent prospective confirmation. The primary, twelve-test secondary family and later descriptive all-pair analysis are distinguished in Supplementary Section S2.

Later descriptive all-pair comparisons place C3 (the Middle-high profile) below Fixed Applied Mean by −0.133 dB (pointwise interval $[-0.170, -0.096]$) and below Late-high by −0.564 dB ($[-0.634, -0.494]$). These are secondary, retrospective comparisons, not the specified timing primary; their complete families and endpoints are reported in Supplementary Section S2. C3 also trails both on LPIPS, FG-SSIM and Edge-SSIM. It improves over Early-high by approximately 0.50 dB. Within these six conditions Fixed High wins FG-PSNR on 172 objects, Late-high on 64, and C3 on none; this is an endpoint-specific count, not a universal quality ranking.

A descriptive four-endpoint comparison complements these winner counts. On 270/300 objects, at least one of the other five controls weakly dominates C3 on all four recorded composite endpoints and improves at least one. Fixed High alone does so on 173 objects, but C3 dominates it on 16; their remaining 111 objects have competing endpoints. This post-inspection, point-estimate analysis is not statistical separation on each object or a ranking of unmeasured material quality. Full counts are in Supplementary Section S2.

The equal-budget comparison identifies an allocation total effect. Evolving residual amplitudes remain trajectory-dependent, preventing a residual-dose-independent mechanism interpretation. Late-high is not universally preferred: Fixed Mean has higher FG-SSIM, and Generic Linear is a strong unmatched-budget practical control. The historical LLH layer/time addendum and complete object distributions remain in Supplementary Section S2.

4.3. Appearance and Color Failure Analysis

The selected color diagnosis contains 26 failure-enriched objects: six earlier figure cases and 20 Fresh C panels. It is not a random sample, a new holdout, or a population estimate. No Adapter, GFL, and LLH were compared on the same inputs, checkpoint, and runner with six target views per object (78 object-condition records). The following paired object-bootstrap intervals are descriptive (20,000 resamples;

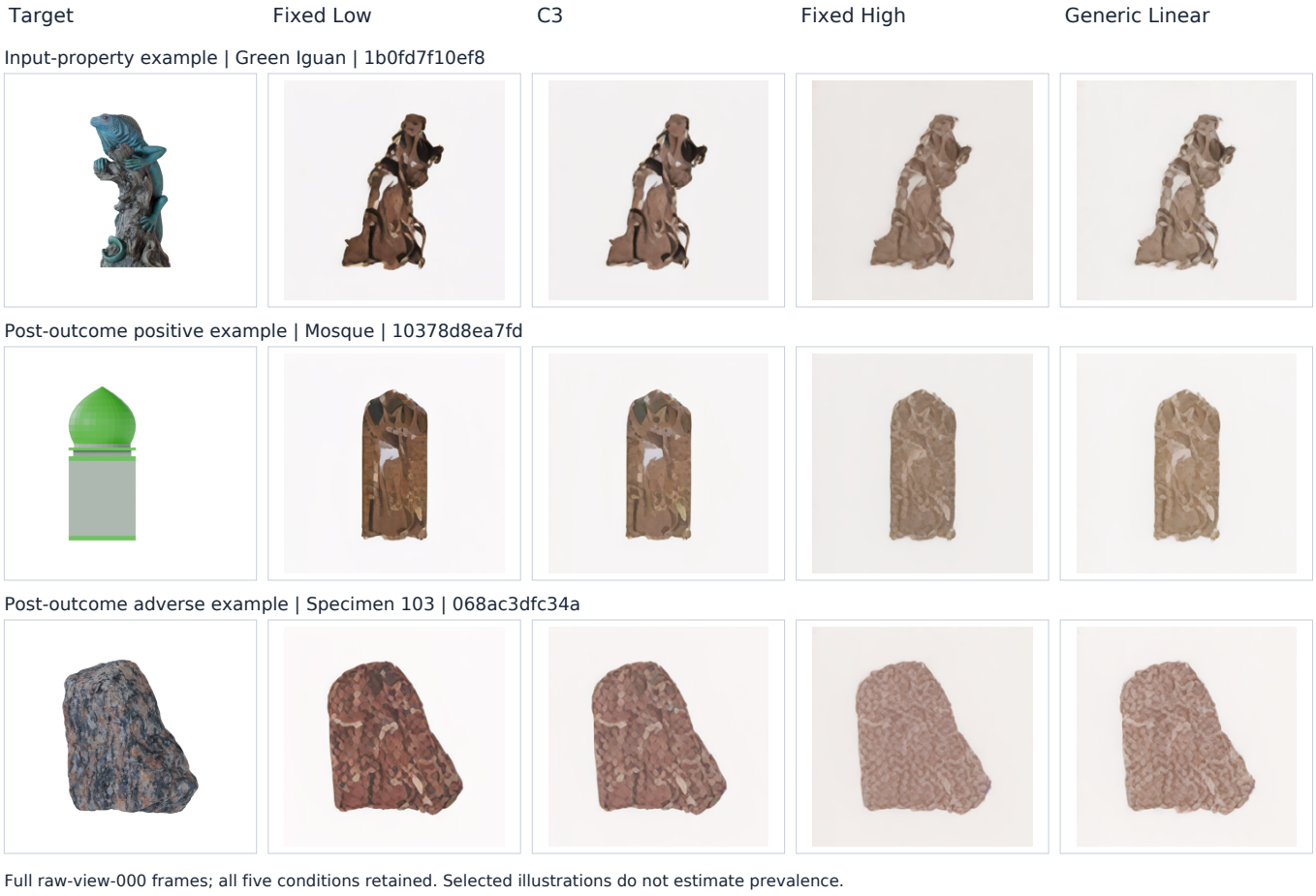


Fig. 2. Fresh C, camera 000. Every row shows Target, Fixed Low, C3, Fixed High and Generic Linear in the same full-frame view. Targets are source RGB conditioning renders, not measured ground-truth albedo. Green Iguan is a colorful, textured organic object selected from source properties. Mosque (+1.736 dB) and Specimen 103 (−0.468 dB) are post-outcome selected positive and adverse C3–Fixed Low FG-PSNR examples. These selected rows illustrate endpoint heterogeneity; they do not estimate prevalence or establish a cohort-wide advantage. Complete source identity and credits are in Supplementary Section S9 and the visual selection manifest.

Table 3. Selected paired color-diagnostic contrasts.

Contrast	ΔE_{00}	95% CI	Favorable
GFL – No Adapter	−19.654	[−26.370, −13.248]	24/26
LLH – GFL	+3.583	[+0.462, +6.712]	–

Failure-enriched $N = 26$; negative differences favor the first condition. Favorable is an object-direction count, not prevalence.

seed 61008), and no confirmatory multiplicity family was defined.

GFL lowered reference-relative CIEDE2000 in 24/26 objects, while visible pink or purple shifts remained in two legacy examples under No Adapter, GFL, and LLH. Their mapping to the original submitted panels cannot be recovered; the Fresh C example in Figure 4 is a different object. The shifts are present in decoded RGB before UV projection or baking. In one case, a ground-truth VAE encode–decode isolate had CIEDE2000 4.086 and $\Delta a^* \approx -1.35$, compared with 22.032 and +1.977 for GFL. This weakens ordinary VAE round-trip error as the sole explanation for that example, without identifying the complete cause.

LLH–GFL increased mean CIEDE2000 while reducing foreground gradient magnitude, high-frequency energy and color entropy. Lower variation therefore did not imply closer reference appearance. The denoising/runtime isolates and three failed correction candidates remain in Supplementary Section S6; they localize observations but do not validate a cause or repair.

Earlier bake results and calibration checks are reported separately in Supplementary Section S5 and are not pooled with the ten-object cohort. The GeoTex residual branch with an unset scale differs from an explicit Constant 1.0 request under the wrapper caps; we label the latter by its executed setting.

Full-object evidence and its selection. The complete views in Figure 2 and the color failure in Figure 4 use a shared full-frame display for every condition within each object; the existing color diagnostic retains its identical pixel crop. Favorable/adverse C3–GFL examples are retained in Supplementary Section S9. Outcome-selected displays illustrate behavior, not prevalence. The historical texture audit does not preserve a traceable denominator or validated texture-loss labels; we therefore do not infer color-failure prevalence or texture loss

Table 4. Ten-object native-UV bake pilot: mean scores over eleven unseen views.

Condition	FG-PSNR	FG-LPIPS	FG-SSIM	CIEDE2000
GT-256 reference bake	22.38	0.0590	0.7683	4.44
Native residual†	11.13	0.1372	0.3376	26.47
No Adapter†	5.92	0.1600	0.3829	40.92
Constant 1.0 (capped)	11.93	0.1354	0.3504	24.18
Fixed Low	12.48	0.1317	0.3835	22.71
Fixed High	12.29	0.1305	0.4587	23.27
Fixed Mean	12.62	0.1303	0.4193	22.32
C3 / Middle-high	12.48	0.1302	0.4059	22.16
Late-high	12.92	0.1277	0.4259	21.65
Generic Linear	11.60	0.1276	0.4458	24.71

† Added exploratory controls. GT-256 is a matched reference bake, not a theoretical bound.

from it. Variation statistics alone cannot establish fidelity.

4.4. Actual Baked Appearance: Local Benefit and External Limits

We selected ten previously unbaked Fresh C objects in UID order using input availability, native UV/material compatibility, triangle count and camera-mask alignment. Only ten of the 300 screened inputs passed these criteria before the cutoff. Their generated-view outputs were known, but their baked outcomes had not been inspected during selection. This compatibility-selected pilot is not a representative sample; five earlier bake objects are reported separately.

Seven conditions share each object's source mesh, reference, seed, six projection cameras, masks, 256-pixel source resolution, atlas resolution, hole filling and eleven unseen evaluation cameras. Original geometry and UVs are retained; unseen cameras do not enter the bake. GT-256 uses the same source resolution, masks and bake parameters, with the same raw atlas coverage as each generated condition. Camera and sampling calibrations are documented in Supplementary Section S5. Evaluation uses unlit base color and does not measure PBR fidelity.

The internally predeclared primary contrast was Late-high minus Fixed Mean on unseen-view FG-LPIPS, proposed from the earlier pilot and tested on the ten newly selected bake objects. Its mean is −0.00265 (95% object-bootstrap CI $[-0.00414, -0.00137]$; exact paired sign-flip $p = 0.00195$), with all ten objects favorable. This is approximately a 2.0% reduction relative to Fixed Mean's mean LPIPS, and identifies a small allocation benefit within this sampling protocol. Generic Linear has a slightly lower mean LPIPS but lower PSNR; Late-high does not dominate every endpoint or control. The matched GT-256 reference bake has mean LPIPS 0.0590, versus 0.1303 for Fixed Mean and 0.1277 for Late-high. A post hoc description therefore puts the absolute gain in context: it closes only 3.7% of the mean Fixed Mean–GT reference gap (pointwise object-bootstrap CI [2.3%,5.3%]). This ratio of paired means is not a theoretical ceiling, a new primary test or evidence of acceptable material fidelity. Leaving out each object in turn keeps the mean Late–Mean difference between −0.00294 and −0.00208, but all substantial reference-appearance discrepancies remain. Supplementary Section S5 supplies the paired values and calculation.

C3–Fixed High has mean PSNR +0.19 dB (95% CI $[-0.65, +0.87]$) and LPIPS −0.00036 ($[-0.00396, +0.00336]$); neither separates under the four-comparison secondary Holm family (both adjusted $p = 1$). Its SSIM is lower by 0.05274 ($[-0.07459, -0.03170]$; Holm $p = 0.00781$). The five-object exploratory C3 PSNR result therefore does not establish a stable benefit in this separate pilot. Late-high is a different allocation from C3.

Across the seven conditions, mean within-object Spearman correlations between generated-view and baked unseen-view rankings are 0.846 for PSNR, 0.804 for LPIPS, 0.746 for SSIM and 0.907 for Edge-SSIM. Broad rankings are positively associated, while the winning condition changes on 7/10 objects for LPIPS (1/10 for PSNR). These are different statements: we do not claim a general reversal of rankings. Figure 3 displays one input-property-selected asset and every ten-object Late-high–Fixed Mean LPIPS effect. The object-level effects range from about −0.00781 to nearly zero; the stone case with the lowest absolute LPIPS improves by only −0.00046. All ten effects favor Late-high on this endpoint, but the visible method differences remain subtle and generated materials differ substantially from the GT self-bake. This compatibility-selected distribution cannot estimate general asset prevalence or establish faithful material reproduction.

Separate twenty-object GSO evaluation. The GSO/MV-Adapter study completes twelve conditions by three seeds on twenty fixed inputs, plus twenty matched GT self-bakes: 740 unique baked meshes and 8,140 unseen-view records. The objects, source meshes, projection cameras, texture resolution and eleven unseen cameras are shared across controls; unseen cameras do not contribute to baking.

The original twenty-object bake protocol failed a discontinuous UV-ramp diagnostic after fifteen objects. A subsequent ground-truth-only calibration addressed finite-resolution mixing across UV boundaries and retained the fixed twenty-object cohort, cameras, masks, materials and resolutions. Because this correction followed inspection of initial outcomes, the resulting GSO evaluation is reported as protocol sensitivity. The original failure and remaining UV/coverage limits are detailed in Supplementary Section S4.

Late-high minus Fixed Mean unseen-view LPIPS is +0.000084 (pointwise interval $[-0.000103, +0.000286]$; original comparison-family Holm $p = 0.788$). This study does not detect the ten-object pilot's benefit. It changes both input domain and conditioning/projection protocol, so it is neither a direct same-protocol replication nor evidence assigning the difference to backbone alone. Table 5 reports the two separate bake primaries; their endpoint remains distinct from the FG-PSNR timing comparisons in Table 7.

Across all twelve GSO conditions, mean within-object image/bake LPIPS-rank correlation is 0.601 (object interval $[0.440, 0.733]$). A post hoc sensitivity restricted to the seven original scale/time controls gives 0.396 ($[0.175, 0.582]$), with the minimum-mean LPIPS condition changing on 11/20 objects. These are positively associated rankings with changing point-estimate winners, not general rank reversal or statistically resolved winner changes. The GT self-bake has mean

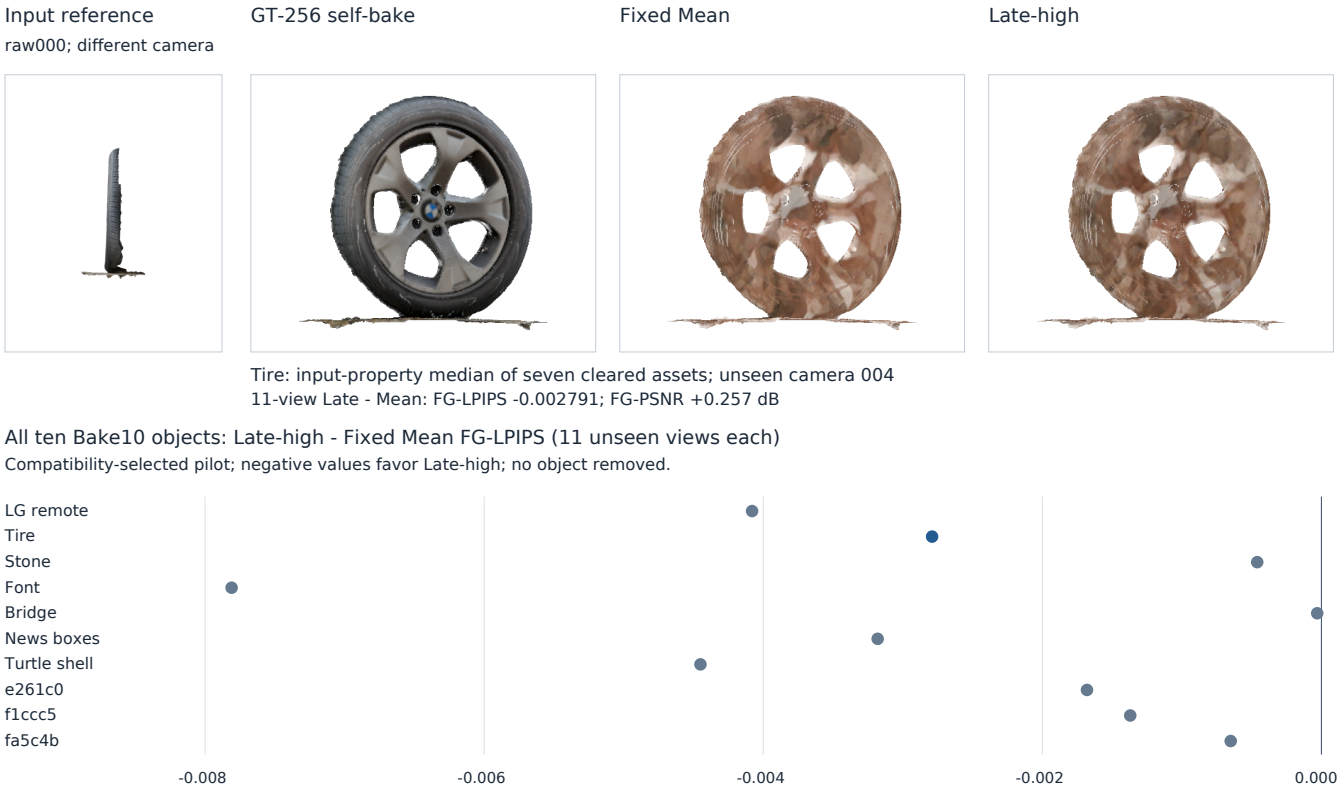


Fig. 3. GeoTex Bake10 asset appearance and object-level allocation effects. The small input inset is raw view 000 and is not pose-matched to the three exported-GLB renders at held-out camera 004. Tire is the median source-texture-complexity input among seven display-cleared assets, not a population-representative object. GT-256, Fixed Mean and Late-high use one shared crop and the same mesh/UVs; all source PNGs are 512×512, without method-specific color edits. The lower plot shows every object in the compatibility-selected ten-object pilot using existing eleven-view Late-high minus Fixed Mean FG-LPIPS means; negative values favor Late-high. It describes heterogeneous, small metric effects, not visible material fidelity or a new inferential test. GT-256 is a reference self-bake, not an upper bound. Full UID values and conditions are in Supplementary Section S5.

Table 5. Unseen-view FG-LPIPS primary contrasts for the two bake studies.

Study (N)	Δ FG-LPIPS	95% CI	p
GeoTex (10)	−0.002652	[−0.004144, −0.001367]	0.00195
GSO (20)	+0.000084	[−0.000103, +0.000286]	0.788

Late-high minus Fixed Mean. GeoTex: exact paired sign-flip. GSO: original-family Holm16-adjusted paired sign-flip. Separate input and calibration protocols are not pooled; non-detection is not equivalence.

LPIPS 0.0602 and PSNR 19.36 dB; average complete-bilinear-footprint coverage is about 78.7%. All foreground remains scored and GT errors are not subtracted. These losses contextualize the small method effects without defining a theoretical quality bound.

Incoming-view consistency. The bake also measures cross-camera color variance at UV texels receiving multiple source views. A constant-black control has zero variance despite its incorrect appearance, showing that this diagnostic measures incoming-view disagreement rather than fidelity. Late-high has a lower mean disagreement than Fixed Mean (−0.002448), while Fixed High has still lower disagreement and higher baked LPIPS. Resampling and overlapping UVs also affect this measure; it does not validate seams or PBR. Full controls and contrasts are in Supplementary Section S5.

Native controls and protocol sensitivity. Subsequently added Native residual and No Adapter outputs on the same ten GeoTex objects are exploratory. They retain matched geometry and GT coverage, and the repeated Constant 1.0 (capped) outputs are bitwise identical. Their apparent PSNR/LPIPS gains for the schedules do not separate under the fifteen-test exploratory family. The complete data are in Supplementary Section S5.

A supporting fifty-object comparison uses a newly trained frozen GeoTex checkpoint with 512-pixel supervision, actual positive/negative reference conditions and CFG 2. Late-high minus Fixed Mean has nonreference PSNR −1.184 dB (interval $[-1.587, -0.849]$) and LPIPS +0.00266 ($[-0.00003, +0.00640]$; adjusted $p = 0.080$). C3 has no corrected advantage over Fixed Mean or Fixed High. This previously studied cohort and changed checkpoint/protocol supply a sensitivity boundary, not an independent backbone confirmation. Training details and full-v29 comparisons are in Supplementary Section S8.

Native-pipeline quality boundary. The recorded full-v29 ControlNet pipeline is a stronger deployment comparator than geometry-disabled output from the new HF-based checkpoint. On the same fifty inputs and five nonreference cameras, the

Table 6. Native-camera MV-Adapter means on fifty input-selected objects.

Condition	FG-PSNR	FG-LPIPS	FG-SSIM	Edge-SSIM
Native / Fixed High	12.60	0.1280	0.5041	0.3526
Fixed Low	12.73	0.1270	0.5066	0.3475
Fixed Mean	12.68	0.1279	0.5068	0.3509
Early-high	12.60	0.1296	0.5041	0.3503
Middle-high	12.71	0.1262	0.5061	0.3470
Late-high	12.74	0.1260	0.5075	0.3514
Generic Linear	12.72	0.1263	0.5072	0.3501

new geometry checkpoint trails full v29 by 2.16 dB FG-PSNR (paired interval $[-3.15, -1.27]$) and has FG-LPIPS higher by 0.0341 ($[0.0235, 0.0464]$); these belong to the separately frozen four-endpoint Holm family. This is a matched-input deployment comparison, not a scale-only comparison: base weights and geometry implementations differ, and the common 50-step, seed-42, CFG-2 setting is not the official command-line default. In an exploratory follow-up that matches the exported v29 base UNet, unscaled geometry improves LPIPS over disabled geometry by 0.0131 yet remains 0.0159 worse than full v29 (paired interval $[0.0085, 0.0237]$). Geometry architecture and training history still differ, and the same previously inspected objects were reused. Thus a local control benefit can coexist with a larger complete-system quality deficit; neither contrast establishes TCAS deployment superiority. Full conditions, tests and grids are in Supplementary Section S8.

4.5. Allocation Responses in Other Conditioning Interfaces

MV-Adapter and MVDiffusion supply different conditioning operators and scale semantics from GeoTex. We evaluate each within its native target/camera contract and do not pool their scores or interpret identical scale labels as common residual doses.

New native MV-Adapter temporal comparison. We selected fifty Fresh C inputs not previously evaluated through MV-Adapter, using input availability and native material/geometry compatibility. Native mesh normalization and six orthographic cameras define both generation and evaluation targets. Three-object identity checks confirmed agreement between the deployed and recorded default paths. An earlier seven-object panel had an invalid camera calibration and is excluded from these comparisons (Supplementary Section S3).

The seven conditions share object inputs, seed, geometry and evaluation code. With 50 steps, low/high are 0.75/1.0; stage-high schedules contain 16 high steps, Fixed Mean is 0.83, and Generic Linear is 0.75–0.91. These five controls have matched per-feature sums within this interface. All 50 objects remain in the primary cohort, including thirteen texture-absent sources with legal constant-white base-color targets. Targets are unlit native-camera base color, not Fresh C's differently normalized reference renders or certified physical albedo.

The internally predeclared primary, Middle-high minus Fixed Mean FG-PSNR, is +0.029 dB (95% object-bootstrap CI $[-0.019, +0.081]$; paired Monte Carlo sign-flip $p = 0.281$; 29/50 favorable). This does not support a stable middle-window gain or establish equivalence. The remaining 104 pair–metric tests use one Holm family. Middle-high has lower Edge-SSIM than Fixed Mean (adjusted $p = 0.00104$); some other placement contrasts separate, including Late-high versus Early-high PSNR. Such endpoint-dependent differences do not demonstrate transferable C3 optimality. Complete contrasts, correction rules and the textured subset are in Supplementary Section S3. This is a native-interface validation of the scheduling question, not architecture-matched replication of GeoTex's capped residual intervention.

External-input native schedule comparison. The source-ordered GSO100 convenience cohort uses the same native MV-Adapter control path and three shared seeds. It includes seven scale/time profiles, with four inherited profiles frozen after earlier baseline exposure. The scheduling comparison is an external protocol sensitivity, not an independent prospective confirmation or parameter search.

The specified all-six-view Middle-high minus Fixed Mean primary is −0.014 dB (interval $[-0.031, +0.004]$; paired $p = 0.122$). A secondary Late-high–Fixed Mean contrast is +0.045 dB ($[+0.024, +0.065]$; Holm83 $p = 0.00335$), with SSIM +0.00134 (adjusted $p = 0.000415$). Its five-nonreference-view LPIPS difference is −0.000086 ($[-0.000551, +0.000413]$; separate Holm21 $p = 1$). Thus a small detected image effect does not establish a perceptual or baked-appearance benefit. Table 7 reports the original Fresh C timing primary first, followed by the two study-specific native-interface FG-PSNR primaries; Table 5 retains the separate bake LPIPS primaries. Full paired secondary families remain in Supplementary Sections S3–S5. The complete rights-cleared native-camera comparison across all seven MV-Adapter conditions is retained in Supplementary Section S3.

Fixed-seed sensitivity. A post-inspection analysis retains all three recorded sampling seeds. GSO100 Late–Mean PSNR averages are positive at each seed (+0.057, +0.038, +0.039 dB), but only 27/100 objects improve at all three and 68/100 have mixed or tied directions. Its nonreference LPIPS averages change direction ($-0.000389, -0.000269, +0.000398$). For the GSO20 bake, only 1/20 objects has lower Late–Mean LPIPS at all three seeds; 14/20 have mixed or tied directions. Sign agreement has no practical-effect threshold. These conditional, object-level summaries do not estimate training randomness or revise the original primary tests; full seed intervals and deltas are in Supplementary Section S4.

The retrospective 75-object MVDiffusion panel scales correspondence-block outputs rather than the GeoTex residual operator. Its layer-wise L–LLH condition is worse than global fixed low in PSNR, structural agreement and reference-color error; LPIPS does not separate. This result and the earlier MV-Adapter panel remain in Supplementary Section S3 and delimit the scope of cross-interface transfer.

Projection-guidance candidates are retained as exploratory negative results in Supplementary Sections S4 and S8; they do not form a validated alternative method.

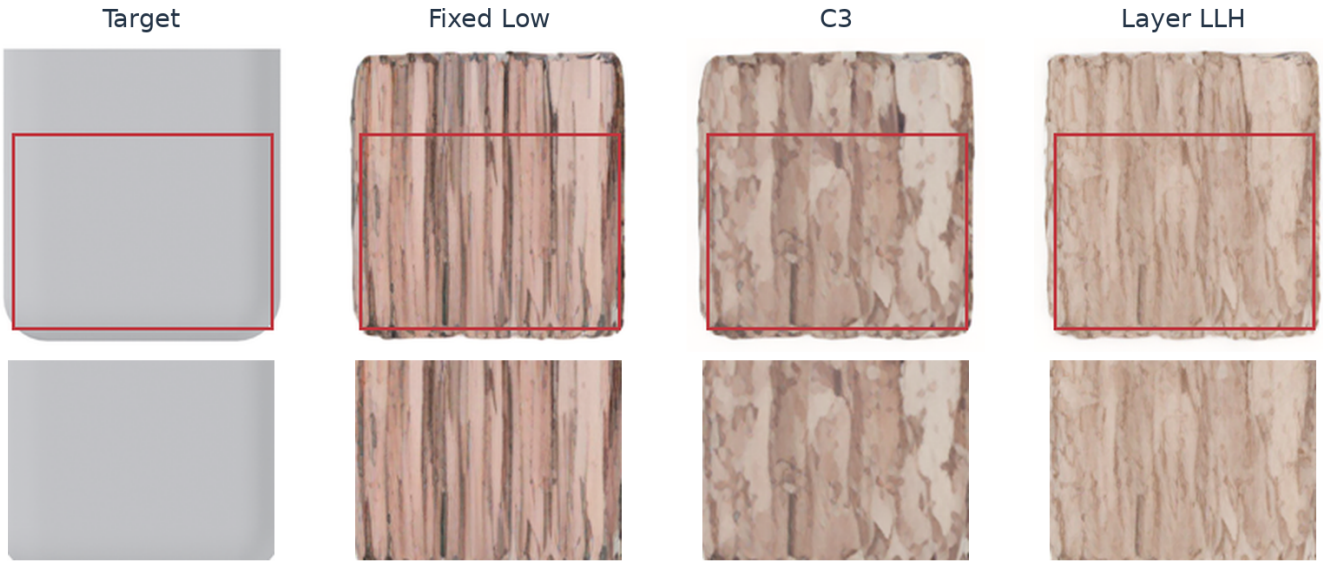


Fig. 4. Fresh C view 0, E5-5-2: Target, Fixed Low, C3 and layer LLH share a complete-object frame and an identical pixel crop. The generated brown pattern does not reproduce the gray target; changing its variation does not establish color fidelity. This previously viewed example came from an existing C3–GFL LPIPS stratum and was not selected for color. It does not reproduce the legacy purple-shift cases. Object credits are supplied in Supplementary Section S9.

Table 7. Specified FG-PSNR schedule primaries.

Study	Contrast	*N*	Δ FG-PSNR (dB)	95% CI (dB)	*p*
GeoTex Fresh C	Middle − mean(Early, Late)	300	−0.033877	[−0.117170, +0.051422]	0.432378
MV-Adapter50	Middle − Fixed Mean	50	+0.028986	[−0.019411, +0.080739]	0.280747
GSO100	Middle − Fixed Mean	100	−0.013660	[−0.030565, +0.003519]	0.122279

Middle/Early/Late denote the corresponding high-window allocations. Each row is its study's specified primary; the GeoTex row is the original timing contrast, not Middle–Fixed Mean. Pointwise paired-object intervals; GeoTex uses 20,000 and the native-interface rows 100,000 paired sign-flip draws. Studies are separate; equal applied scalar sums do not imply equal realized residual dose.

4.6. Historical Cohort Sensitivity and Prior Evaluation

The 276-slot retraining tests endpoint sensitivity on a previously exposed historical request list. Seventy-nine requested UIDs overlapped the earlier recorded training list and were removed, leaving 1,039 training UIDs with no requested-list overlap. During evaluation, an empty-input error at requested slot 46 triggered an inherited loader's random replacement whose actual UID was not recorded. The 1,104 rows therefore represent 276 requested slots under four paired conditions, not verified coverage of 276 distinct held-out objects. This panel is reported as a protocol/checkpoint sensitivity analysis; it does not recover the historical main checkpoint or reproduce its reported +0.96 dB result.

C3 minus Fixed High is +1.699 dB in full-frame PSNR (276/276 slots positive), but −1.314 dB in FG-PSNR (9/276 favorable). FG-LPIPS is higher by 0.01127, FG-SSIM lower by 0.05961, and Edge-SSIM lower by 0.02777. Excluding the known failed requested slot preserves the opposing PSNR means (+1.697 and −1.311 dB) and the LPIPS loss (+0.01127). White-background normalization also favors Fixed High. The raw full-frame gain therefore does not support reference-texture fidelity. C3 improves FG-PSNR over Fixed Low by 0.783 dB, while their descriptive FG-LPIPS interval includes zero. Slot-bootstrap diagnostics do not establish independent-object inference because the substituted UID is unknown. Fixed Mean, moved high windows and Generic Linear were not evaluated here, so this panel cannot test temporal allocation. The requested-slot condition means (Supplementary Table S36) and paired endpoint contrasts (Supplementary Table S35) remain in Section S7, with the unknown actual-UID substitution and technical, non-confirmatory scope stated alongside.

The original blinded 3AFC study on thirty objects and twenty-four participants reports C3's overall-quality share as 58.1% (two-level participant/object-bootstrap interval [50.1, 65.8]%). This preference applies to the original stimuli; it does not measure texture fidelity for the retrained checkpoint or unseen baked appearance. Counts and endpoint-specific results are retained in Supplementary Section S7.

Historical learned-control comparison. Previously reported Full Adaptive Correction (FAC) configurations all reduced FG-PSNR relative to TCAS in their stated controlled re-examination. The original training pool was reported as 1,706 objects, while a recovered configuration refers to 1,200; seed, checkpoint and sampled-run records are incomplete. This result is retained as historical context and does not establish a general comparison between learned and inference-time control. Source identity, recovered code and the limits of training

replay are detailed in Supplementary Section S7.

5. Discussion and Limitations

Strength and timing answer different questions. A low–high–low curve can appear effective when the fixed comparator is weak, yet that contrast changes both amplitude and placement. Fresh C makes this distinction concrete: stronger competitive controls change the practical recommendation, and the preserved equal-budget primary does not support the earlier middle-window advantage. Separate pairwise results favor Fixed Mean and Late-high over C3 on several endpoints, while Late-high trades lower LPIPS and higher PSNR against lower FG-SSIM than Fixed Mean. These observations change the tested recommendation, but do not show that every temporal schedule is ineffective or that Late-high is universally best.

Matching scalar sums does not identify a residual-dose mechanism. The native shallow cap makes all three C3 requests identical after clipping in that group. A requested-profile comparison can therefore describe an intervention that was never applied as drawn. Reporting the implemented per-group profiles and sums removes that ambiguity. Matching those sums still leaves the adapter activation and hidden-state trajectory free to change. The matched-budget contrasts estimate the total effect of allocating the scalar control, including the resulting correction changes; they do not isolate timing at an equal realized residual dose. Raw correction norms also depend on tensor dimensions. The descriptive activation traces in Supplementary Section S1 cannot validate a universal stage mechanism or assign a structural/texture function to each cap group.

What baking adds to the interpretation. Generated-view metrics are informative: broad within-object rankings are positively associated with baked rankings. However, projection and texture construction add losses, and the minimum-score control can change between image and asset endpoints. The association weakens in GSO when restricted to closely competing scale/time controls. These are associations and changes of point-estimate winners, not statistically resolved rank reversals.

In the compatible GeoTex pilot, Late-high reduces LPIPS by about 2% relative to Fixed Mean but closes only 3.7% of its gap to the matched GT bake. The latter is a reference for projection/resampling loss, not a theoretical ceiling; GT error is not subtracted from method errors. Reference-color and detail discrepancies remain visible, and the separate GSO protocol does not detect the same allocation benefit. Thus the asset evidence quantifies a local control effect alongside the much larger remaining appearance problem. Incoming-view disagreement is also a distinct diagnostic: a constant wrong color can have zero variance. Neither this diagnostic nor unlit LPIPS certifies seams, physical albedo or PBR/material fidelity.

What can and cannot transfer. The native MV-Adapter50 and GSO100 primary intervals do not detect a middle-window advantage; they do not establish equivalence. The secondary GSO late-window PSNR effect is small and has no detected non-reference LPIPS counterpart. A reference-conditioned GeoTex checkpoint gives a different Late–Mean direction, and MVDiffusion tests only a partial correspondence operator. These observations limit a portable middle-stage rule. Because inputs, weights, operators and sampling/projection protocols change together, the studies cannot attribute their response differences to a backbone or domain alone. Equal numerical scale labels across them are not common physical doses.

Observed input heterogeneity. In the separate retrospective FRESH_CONFIRM_300 cohort, LLH minus Fixed Low changes with reference-image texture variation: foreground Laplacian variance has Spearman $\rho = -0.524$ with the FG-PSNR effect, and the four 75-object variance quartiles have mean effects +4.140, +1.127, −0.087 and −0.029 dB. This cohort uses its own frozen inputs and LLH/GFL comparison; it is not the Fresh C matched-budget primary. The association suggests an input dimension worth measuring when interpreting control responses, but does not identify a causal texture effect, a threshold, or a validated adaptive schedule. The complete ten-test retrospective family is in Supplementary Section S7.

Consequences for evaluating a texturing control. A practical comparison should first retain the deployed input and camera contract and compare native/default and competitive fixed settings. To interpret placement, it should report the cap-adjusted profiles and match per-group applied sums, with residual traces used to describe rather than certify dose. Reference-color errors and full-object displays should accompany image metrics. Actual baking with shared meshes, unseen cameras and a matched reference bake can then assess whether the effect matters for the asset. Finally, fixed-seed differences should be examined at object level: positive cohort means can coexist with mixed directions across seeds. This is an evaluation sequence supported by the observed failure of stronger interpretations, not a newly validated selection algorithm. The evidence gives no general reason to replace a deployment's fixed setting with C3.

Selection, statistics and replication limits. The bake cohorts favor compatible native-UV assets and GSO uses a source-ordered convenience sample. The two bake studies have different conditioning/projection protocols; the revised GSO calibration followed inspection of initial outcomes. Equal-budget analyses were specified on an already evaluated Fresh C cohort and remain retrospective; the dominance, seed and coverage addenda are descriptive. Separate multiplicity families and pointwise intervals are not one overall error guarantee. Views and seeds remain nested within object, and three fixed seeds do not estimate training-checkpoint variation.

Fresh C uses controlled retraining and sampling rather than the authenticated historical main checkpoint or the complete default pipeline. The original +0.96 dB result is not reproduced, and the historical-request panel has an unknown substituted input, precluding a verified 276-object holdout interpretation. Original preference evidence concerns its own stimuli; the historical FAC manifest and weights are incomplete. These limits are retained in Supplementary Sections S6–S8. They constrain historical replication and deployment claims while leaving the paired within-protocol comparisons interpretable.

6. Conclusion

Geometry-control allocation in multi-view texturing must be interpreted at the implemented interface and the final asset endpoint. On the held-out GeoTex cohort, competitive baselines and cap-adjusted budgets change the interpretation of C3's gain over Fixed Low. The specified matched-budget primary does not support a middle-window advantage, while separate pairwise contrasts reveal endpoint-dependent placement effects. Native-UV baking measures a small, heterogeneous Late-high benefit on ten compatible objects together with persistent appearance error; a separate GSO study does not detect this benefit. A full-native comparator remains substantially stronger than the new geometry checkpoint under a recorded matched-input protocol, and native MV-Adapter comparisons do not establish the proposed middle-window gain. The reusable finding is a comparison chain: identify the applied control, retain strong within-interface alternatives, examine object-dependent responses, and measure the textured asset against a reference bake and complete-system boundary. This supports better evaluation decisions without a universal schedule, independently derived stage mechanism or faithful material reproduction.

Data availability

The accompanying research package contains object-level metrics, frozen protocols and configuration records, analysis scripts, and the LaTeX source needed to rebuild the manuscript and Supplement. Recovered FAC source is supplied with a record of its entry points and validation scope; these component checks do not constitute a historical full-model replay. Statistical summaries can be recomputed without model weights, while fresh inference requires the recorded weights, runtime and licensed source assets. Raw generated-image archives, trained checkpoints and participant-level responses are not included. Source assets must be acquired under their recorded licenses. Additional analysis records may be requested from the corresponding author, subject to licensing, privacy and archive constraints.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Liu R, Wu R, Van Hoorick B, et al. Zero-1-to-3: Zero-shot one image to 3D object. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

[2] Shi R, Chen H, Zhang Z, et al. Zero123++: A single image to consistent multi-view diffusion base model. arXiv:2310.15110, 2023.

[3] Liu Y, Lin C, Zeng Z, et al. SyncDreamer: Generating multiview-consistent images from a single-view image. *International Conference on Learning Representations*, 2024.

[4] Long X, Guo Y C, Lin C, et al. Wonder3D: Single image to 3D using cross-domain diffusion. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[5] Shi Y, Wang P, Ye J, et al. MVDream: Multi-view diffusion for 3D generation. *International Conference on Learning Representations*, 2024.

[6] Wang P, et al. ImageDream: Image-prompt multi-view diffusion for 3D generation. arXiv:2312.02201, 2023.

[7] Xu J, et al. InstantMesh: Efficient 3D mesh generation from a single image with sparse-view large reconstruction models. *Advances in Neural Information Processing Systems*, 2024.

[8] Li P, Liu Y, Long X, et al. Era3D: High-resolution multiview diffusion using efficient row-wise attention. *Advances in Neural Information Processing Systems*, 2024.

[9] Chen D Z, et al. Text2Tex: Text-driven texture synthesis via diffusion models. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

[10] Richardson E, Metzer G, Alaluf Y, et al. TEXTure: Text-guided texturing of 3D shapes. *ACM SIGGRAPH 2023 Conference Proceedings*, 2023.

[11] Poole B, Jain A, Barron J T, Mildenhall B. DreamFusion: Text-to-3D using 2D diffusion. *International Conference on Learning Representations*, 2023.

[12] Chen R, Chen Y, Jiao N, Jia K. Fantasia3D: Disentangling geometry and appearance for high-quality text-to-3D content creation. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

[13] Shao M, Xiong F, Sun Z, Xu M. MVPainter: Accurate and detailed 3D texture generation via multi-view diffusion with geometric control. arXiv:2505.12635, 2025.

[14] Kynkäänniemi T, Aittala M, Karras T, et al. Applying guidance in a limited interval improves sample and distribution quality in diffusion models. *Advances in Neural Information Processing Systems*, 2024.

[15] Perla S R K, Zhang H, Mahdavi-Amiri A. Advances in neural 3D mesh texturing: A survey. *Computer Graphics Forum*, 2026, e70392.

[16] Guzmán J E, Mould D, Paquette E. Adaptive multiresolution exemplar-based texture synthesis on animated fluids. *Computers & Graphics*, 2026, 134: 104490.

[17] Lv X, Wu Z, Xu J, et al. Interpretable procedural material graph generation via diffusion models from reference images. *The Visual Computer*, 2025, 41(13): 11195–11205.

[18] Zeng X, Chen X, Qi Z, et al. Paint3D: Paint Anything 3D with Lighting-Less Texture Diffusion Models. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[19] Cheng W, Mu J, Zeng X, et al. MVPaint: Synchronized multi-view diffusion for painting anything 3D. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

[20] Zhao Z, Lai Z, Lin Q, et al. Hunyuan3D 2.0: Scaling diffusion models for high resolution textured 3D assets generation. arXiv:2501.12202, 2025.

[21] Huang Z, et al. MV-Adapter: Multi-view consistent image generation made easy. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025.

[22] Tang S, Zhang F, Chen J, Wang P, Furukawa Y. MVDiffusion: Enabling holistic multi-view image generation with correspondence-aware diffusion. *Advances in Neural Information Processing Systems*, 2023.

[23] Liu Y X, Xie M S, Liu H Y, Wong T T. Text-guided texturing by synchronized multi-view diffusion. *SIGGRAPH Asia 2024 Conference Papers*, 2024.

[24] Xu Q, Han X G, Zhang L. WonderTex: Consistent-and-seamless texture generation with text-guided multi-view image diffusion models. *IEEE Transactions on Visualization and Computer Graphics*, 2026, 32(7): 5815–5825.

[25] Mou C, Wang X, Xie L, et al. T2I-Adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.

[26] Zhang L, Rao A, Agrawala M. Adding conditional control to text-to-image diffusion models. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

[27] Ye H, Zhang J, Liu S, Han X, Yang W. IP-Adapter: Text compatible image prompt adapter for text-to-image diffusion models. arXiv:2308.06721, 2023.

[28] Hu E J, Shen Y, Wallis P, et al. LoRA: Low-rank adaptation of large language models. *International Conference on Learning Representations*, 2022.

[29] Liu S Y, Wang C Y, Yin H, et al. DoRA: Weight-decomposed low-rank adaptation. *International Conference on Machine Learning*, 2024.

[30] Zhang Q, Zuo S, Liang C, et al. AdaLoRA: Adaptive budget allocation for parameter-efficient fine-tuning. *International Conference on Learning Representations*, 2023.

[31] Kulkarni A S. Scheduled Style Injection: Expanding the Style-Content Pareto Frontier in Training-Free Diffusion-Based Style Transfer. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), NTIRE Workshop*, 2026.

[32] Ho J, Jain A, Abbeel P. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 2020.

[33] Song J, Meng C, Ermon S. Denoising diffusion implicit models. *International Conference on Learning Representations*, 2021.
[34] Ho J, Salimans T. Classifier-free diffusion guidance. *NeurIPS Workshop on Deep Generative Models and Downstream Applications*, 2021. arXiv:2207.12598.
[35] Castillo A, Kohler J, Pérez J C, et al. Adaptive guidance: Training-free acceleration of conditional diffusion models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.
[36] Wang Z, Bovik A C, Sheikh H R, Simoncelli E P. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 2004, 13(4): 600–612.
[37] Zhang R, Isola P, Efros A A, et al. The unreasonable effectiveness of deep features as a perceptual metric. *CVPR*, 2018.
[38] Sadat S, Hilliges O, Weber R M. Eliminating oversaturation and artifacts of high guidance scales in diffusion models. *International Conference on Learning Representations*, 2025.
[39] Chung J, Hyun S, Heo J P. Style Injection in Diffusion: A Training-free Approach for Adapting Large-scale Diffusion Models for Style Transfer. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
[40] ControlNet project. Official depth-to-image demo: layer-dependent control scales. Source code, fixed commit f2e1bc38dc8d925fb542be4c138c46013ace1988, 2023. `https://github.com/lllyasviel/ControlNet/blob/f2e1bc38dc8d925fb542be4c138c46013ace1988/gradio_depth2image.py`. Accessed 12 October 2026.
[41] Hugging Face. Diffusers v0.21.0: ControlNet conditioning scale and guidance start/end. Official versioned documentation, 2023. `https://huggingface.co/docs/diffusers/v0.21.0/api/pipelines/controlnet`. Accessed 12 October 2026.